

Overview

In this assignment, you will create a **comprehensive AWS Service Selection Guide** that helps practitioners make informed decisions between commonly-confused AWS services. You will select **TWO service sets** from the provided options and develop detailed guidance that goes beyond simple feature comparisons to address real-world architectural decision-making.

This is not a research paper—it's a **practical reference guide** that a cloud architect or developer could actually use when designing systems. Your guide should reflect deep understanding of service capabilities, limitations, costs, and appropriate use cases.

Instructions

Service Set Selection

Choose **TWO** of the following service sets to analyze in depth:

Set 1: Compute Services

Lambda vs. Fargate vs. EC2 vs. Lightsail

Set 2: Database Services

RDS vs. DynamoDB vs. Aurora vs. Redshift

Set 3: Storage Services

S3 vs. EFS vs. EBS vs. Storage Gateway

Set 4: Integration & Messaging Services

SQS vs. SNS vs. EventBridge vs. Kinesis

Set 5: Application Delivery & Routing

API Gateway vs. Application Load Balancer vs. CloudFront vs. Route 53

Required Deliverables

For **EACH** of your two selected service sets, create:

1. Service Overview Section (Required)

For each service in the set, provide:

- **Purpose Statement:** 2-3 sentence description of what the service does
- **Primary Use Cases:** 3-5 bullet points of ideal scenarios
- **Key Capabilities:** Notable features that differentiate it from alternatives
- **Pricing Model Summary:** How costs are calculated (per hour, per request, per GB, etc.)

2. Decision Criteria Framework (Required)

Develop a structured framework with **at least 6 decision criteria** relevant to your service set. Examples might include:

- **Workload characteristics** (predictable vs. variable, long-running vs. short-lived)
- **Data structure requirements** (structured, semi-structured, unstructured)
- **Scalability patterns** (horizontal, vertical, auto-scaling)
- **Performance requirements** (latency, throughput, IOPS)
- **Cost considerations** (fixed vs. variable, cost at scale)
- **Operational complexity** (managed vs. self-managed)
- **Integration requirements** (VPC, other AWS services, on-premises)

For each criterion, explain:

- Why it matters for this decision
- How each service performs against this criterion
- When this criterion should be weighted more heavily

3. Comparative Analysis (Required)

Create a **comparison table or matrix** that shows how each service performs across your decision criteria. Use a clear rating system (e.g., Excellent/Good/Fair/Poor or 1-5 scale) with brief justifications for each rating.

Example structure:

Criterion	Service A	Service B	Service C	Service D
Cost at Low Scale	Good - Pay per use	Excellent - Free tier	Fair - Fixed costs	Poor - High minimum

Note: Include brief justifications for each rating to help readers understand the trade-offs.

4. Use Case Scenarios (Required)

Develop **2 use case scenarios** that demonstrate your decision-making process. For each scenario, provide:

- **Scenario Description:** Brief overview of the business/technical need
- **Key Requirements:** The 2-3 most important decision criteria for this scenario
- **Service Selection:** Which service you recommend
- **Rationale:** How you applied your decision criteria to reach this conclusion

Example:

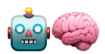
Scenario: E-commerce Product Catalog

A retail company needs to store product information including names, descriptions, prices, and inventory counts. They expect 10,000 products initially, growing to 100,000 within two years. The application needs to support complex queries like "find all products in category X with price < \$Y."

Key Requirements: Structured data with relationships, complex query support, predictable read/write patterns

Service Selection: RDS (MySQL or PostgreSQL)

Rationale: The relational nature of product data (categories, suppliers, inventory) and need for complex SQL queries makes RDS the clear choice. DynamoDB would require complex query workarounds, while Aurora would be over-engineered for this scale. Redshift is designed for analytics, not transactional workloads.



AI Usage Guidelines

AI is PERMITTED but not REQUIRED for this assignment.

You may use AI tools (ChatGPT, Claude, Gemini, etc.) to:

- Research AWS service capabilities
- Generate comparison tables or initial drafts
- Help structure your thinking
- Check technical accuracy
- Improve writing clarity

You may **NOT**:

- Upload this entire document into an AI tool
- Submit AI-generated content without verification and personalization

- Use AI to replace your own critical thinking and analysis
- Copy AWS documentation verbatim (with or without AI)

Important: Generic or superficial analysis will be evident regardless of AI use. The depth of understanding demonstrated in your scenarios and decision frameworks cannot be faked with AI alone—it requires synthesis of course concepts and hands-on knowledge.

Submission

Submit your document in Brightspace.

Evaluation Preview

Your work will be evaluated on:

- **Technical Accuracy** (35%): Correct information about service capabilities, limitations, and use cases
- **Decision Framework Quality** (30%): Usefulness and clarity of decision criteria and comparison tables
- **Scenario Analysis** (20%): Realism and clarity of use case scenarios with proper justification
- **Professional Quality** (15%): Writing clarity, document organization, formatting, and polish

See the detailed rubric for specific performance descriptors at each level.